S.S.V.P.S's B. S. Deore Polytechnic, Dhule

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SEMINAR AND PROJECT INITIATION COURSE

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Topic : Smart Bill Book

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Abstract

The Smart Bill Book – Gas Agency Management System* is a web-based application developed using Python, Streamlit, and SQLite to simplify and automate the daily operations of LPG gas agencies. The system manages customer registration, LPG connection requests, cylinder bookings, billing, invoice generation, and delivery tracking.

It provides separate dashboards for administrators, employees, and customers to manage their activities efficiently. The application reduces paperwork, minimizes manual errors, improves data accuracy, and saves time. SQLite is used to store and manage customer details, booking records, billing information, and delivery status.

The main aim of this project is to provide a user-friendly, reliable, and efficient digital solution for gas agency management..

1. Introduction

- The Smart Bill Book is a web-based Gas Agency Management System designed to simplify LPG gas agency operations.
- It is developed using Python, Streamlit, and SQLite.
- The system manages customer registration and LPG connection requests.
- It provides facilities for cylinder booking and delivery tracking.
- The application automates billing and invoice generation.
- It maintains customer details, booking records, and payment information.
- The main aim is to reduce paperwork, minimize errors, and improve the efficiency of gas agency management.

2. Literature Survey

Sr. No.	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
1	Computerized Gas Agency Management System	DBMS, Application Software	Customer, Stock, Billing Records	Computerized billing and stock management	Accurate data entry required
2	Automated Gas Agency Billing System	Database, Billing Software	Customers, Cylinder Prices, Bills	Simplify billing and reduce errors	Incorrect data causes billing errors
3	Gas Booking and Customer Management System	Web Application, Database	Customers, Bookings, Delivery Dates	Manage customers and bookings	Regular updates required

Sr. No.	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
4	Online Gas Booking and Payment Integration	Web Technologies, Database, Payment Gateway	Bookings, Payments, Stock	Online booking and payment	Internet and payment security required
5	Inventory and Stock Management System	DBMS, Inventory Software	Cylinder Stock, Quantities	Track stock and availability	Incorrect stock entries
6	Customer Database Management System	Relational database, SQL	Customer IDs, names, addresses and contact information	To store customer information systematically and retrieve records quickly	Requires access control, data validation and database backups

Sr. No.	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
7	Digital Invoice Generation System	Billing Software, Database	Invoice Numbers, Prices, Totals	Generate digital invoices	Incorrect pricing
8	Transaction Management System	Database, SQL	Payments, Dates, Bills	Maintain transaction history	Duplicate or incomplete records
9	Web-Based Gas Agency Management System	HTML, CSS, JavaScript, Database	Customers, Bookings, Stock, Billing	Centralized agency management	Security and maintenance required

Sr. No.	Approach Type	Technologies Used	Dataset	Key Objective	Limitations
10	Automated Bill Calculation System	Programming, Database, Calculation Logic	Price, Quantity, Charges	Automatic bill calculation	Incorrect input handling
11	Billing Report and Record Management System	Database, Reporting Tools	Daily Bills, Monthly Transactions	Generate billing reports	Accurate data required
12	Smart Bill Book for Gas Agency	Web Technologies, Database, Billing Logic	Customers, Bills, Payments, Bookings	Digital billing and record management	Testing and maintenance required

3. Information to seminar topic

- Technology: Python, Streamlit, and SQLite
- Main Purpose: To simplify and automate the daily operations of an LPG gas agency.
- Key Concept: Managing customer details, cylinder bookings, billing, and delivery records digitally.
- Approach: Provides a web-based system with separate dashboards for customers, administrators, and employees.
- Main Benefit: Reduces paperwork, minimizes manual errors, and saves time.
- Application: Can be used by LPG gas agencies to manage customers, bookings, invoices, and delivery status efficiently.

4. Advantages / Disadvantages

Advantages

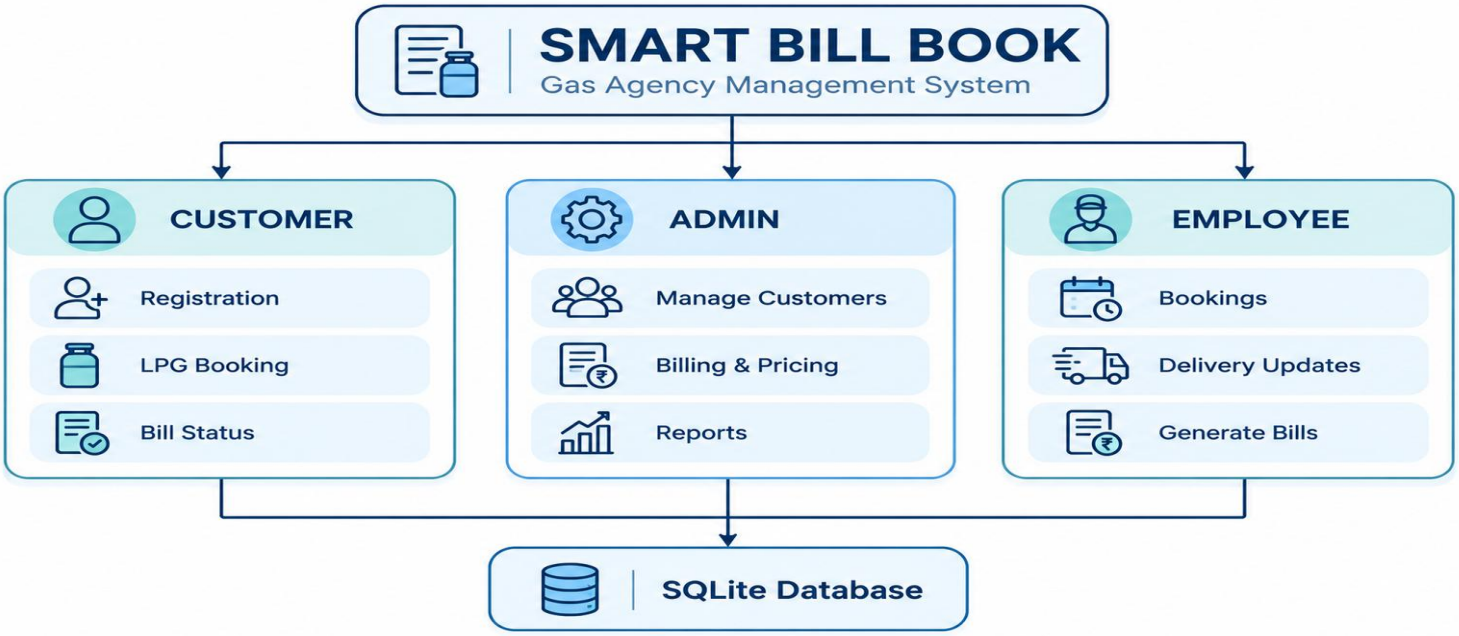
- Protects user's personal information.
- Reduces unnecessary sharing of sensitive data.
- Keeps important information needed for AI responses.
- Helps maintain the quality of AI-generated answers.
- Makes AI usage safer and more trustworthy.

Disadvantages

- May incorrectly identify personal information.
- Could hide details that are important for a response.
- Adds extra processing time before sending a prompt.
- Requires regular updates to improve detection accuracy.
- Cannot guarantee complete protection against every privacy risk.

5. Conclusion

The Smart Bill Book is a web-based Gas Agency Management System developed using Python, Streamlit, and SQLite. It simplifies customer registration, LPG cylinder booking, billing, and delivery management. The system provides separate dashboards for customers, administrators, and employees. It reduces manual paperwork, minimizes errors, and maintains accurate records. Overall, the system improves the efficiency, accuracy, and reliability of gas agency operations.



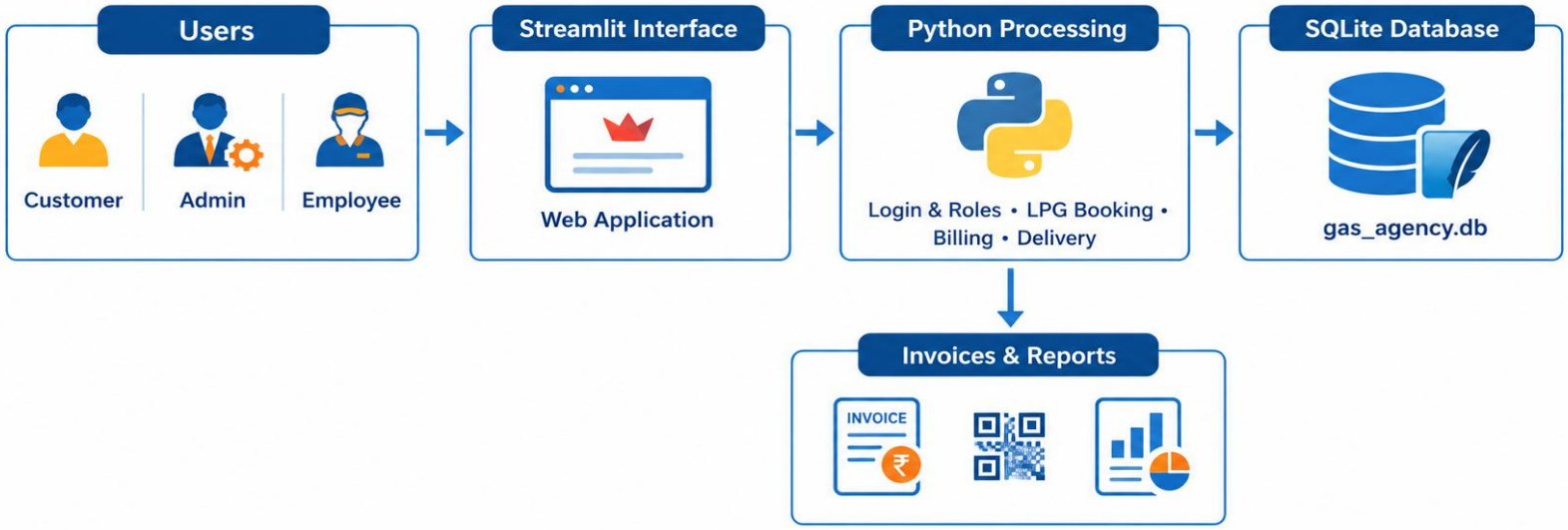
6. Project Initiation

The Smart Bill Book project was initiated to address the problems of manual billing, customer record management, and LPG cylinder booking in gas agencies. Traditional methods require paperwork and may cause calculation errors, delays, and difficulties in maintaining accurate records.

The project aims to develop a web-based Gas Agency Management System using Python, Streamlit, and SQLite. It automates customer registration, LPG booking, billing, invoice generation, and delivery tracking. The system provides separate dashboards for customers, administrators, and employees to manage their activities efficiently. As a result, it reduces paperwork, minimizes errors, saves time, and improves the overall efficiency of gas agency operations.

6. 1 Description of problem statement. identification.

The traditional gas agency system involves manual customer registration, LPG booking, billing, and delivery tracking, which can cause errors and delays. The Smart Bill Book system automates these activities through a web-based application. It uses Python, Streamlit, and SQLite to manage customer details, bookings, bills, and invoices. The system maintains organized records and reduces paperwork. It improves accuracy, saves time, and increases the efficiency of gas agency operations.



6. 2 Scope and objectives.

Objectives

- Automate customer registration and LPG cylinder booking.
- Generate accurate bills and invoices.
- Manage customer details and booking records.
- Track cylinder delivery and payment status.
- Reduce manual work and calculation errors.
- Improve the efficiency of gas agency operations.

Scope

- Develop a web-based Gas Agency Management System.
- Use Python, Streamlit, and SQLite for development.
- Provide separate access for customers, administrators, and employees.
- Maintain customer, booking, billing, and payment records.
- Generate invoices and support delivery tracking.
- Simplify gas agency operations through centralized data management.

6. 3 Stakeholder

- **End Users:** Individuals that will use the AI-powered tools and want to make sure their personal data is kept safe.
- **AI Service Providers:** The companies offering services of AI chatbots and large language models.
- **Developers:** The individuals that will design the AI-based Privacy Gateway.
- **Organizations:** The businesses that will be utilizing the AI tools and have concerns about the security of their data.
- **Privacy and Security Teams:** The personnel in charge of the well-being of the personal data within the organizations.
- **Researchers:** The individuals that will conduct research on the AI privacy tools and enhance their quality.

6. 4 Seminar Execution Plan

1. **Current Scenario:** Manual management of customer details, LPG bookings, billing, and deliveries.

2. **Privacy Challenges:** Billing errors, service delays, and difficulty maintaining records.

3. **Existing Approaches:** Paper registers and basic software for gas agency operations.

4. **Research Gap:** Lack of centralized data management and automated billing.

5. **Research:** Develop Smart Bill Book using Python, Streamlit, and SQLite.

6. **Future Impacts:** Reduce paperwork, improve accuracy, and provide faster service.

7. **Future Work:** Introduce online payments, SMS notifications, and real-time delivery tracking.

6.5 Platform/ Equipment/ Resources

Hardware Requirements:
- Laptop or desktop computer.
- Minimum 4 GB RAM (8 GB recommended).
- Stable internet connection.

Software Requirements:
- Operating System: Windows or Linux.
- Programming Language: Python.
- Framework: Streamlit.
- Database: SQLite.
- Code Editor: Visual Studio Code.

Other Resources:
- Customer details and LPG booking data.
- Billing and invoice generation modules.
- SQLite database for storing records.
- Testing tools to verify system functionality.

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Thank you